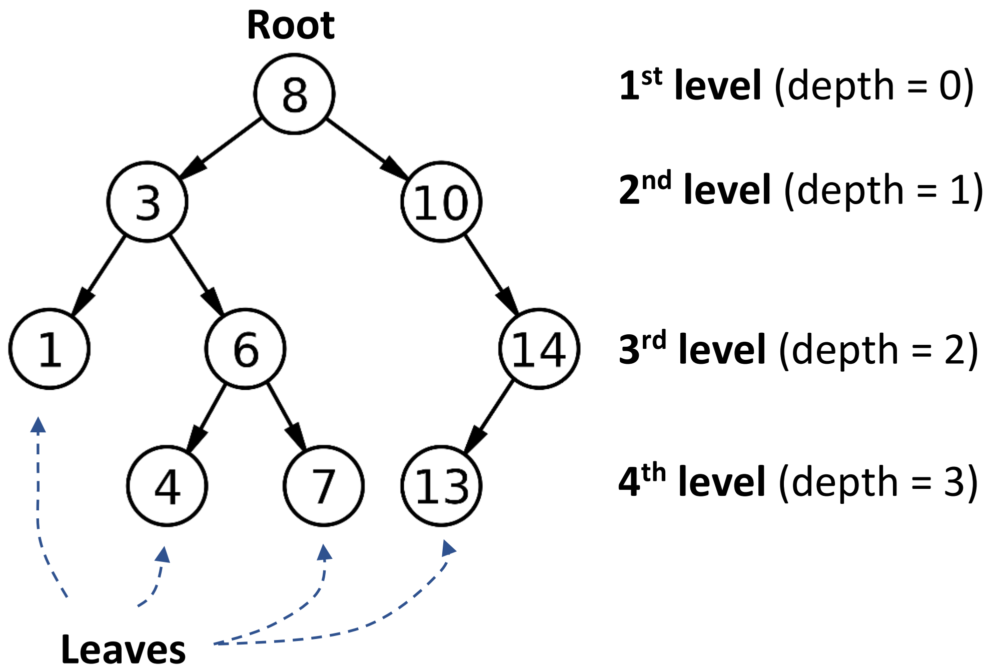
Binary Search Tree Terms

A **binary tree** is a tree data structure in which each node has at most two children, which are referred to as the left child and the right child.



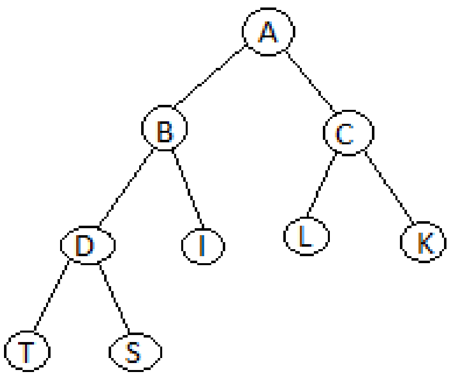
Nodes are either leaves or else they are internal. The nodes storing 3, 6, 8, 10, and 14 are internal, and those storing 1, 4, 7, and 13 are leaves.

root	the node from which all other nodes "descend".
leaf	a node with no children.
parent	A node which has children. Also known as an **internal node**.
child	A node which has a parent. The only node that isn't a child is the root.
subtree	A node *n* and all its descendants. Node *n* is called the root of the subtree.
depth of a node	the number of **edges** from the node to the root.
height of a node	the number of **edges** from node to the lowest leaf in the subtree that has this node as its root.
height / depth of the tree	the length of the path from the root to the bottom node.
diameter	the number of **edges** on the longest path between any two nodes in the tree. • Does necessarily go through the root!
width	the width of a tree is the largest number of **nodes** on any level

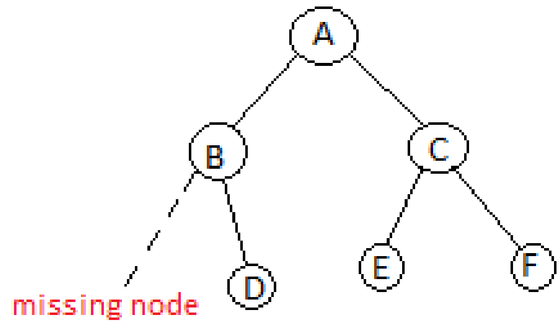
Special Forms of Binary Trees

Complete: In a complete tree, every level above the bottom level is filled.
The bottom level has all nodes shifted as far left as possible.

Complete

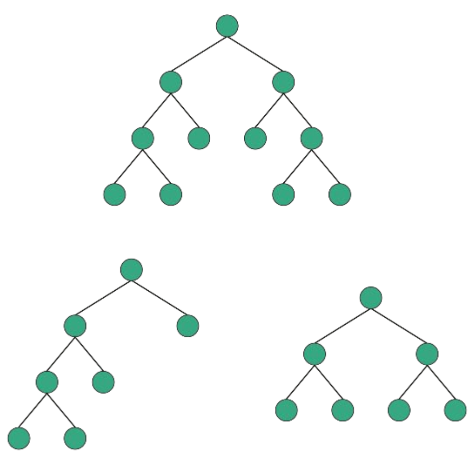


In-complete

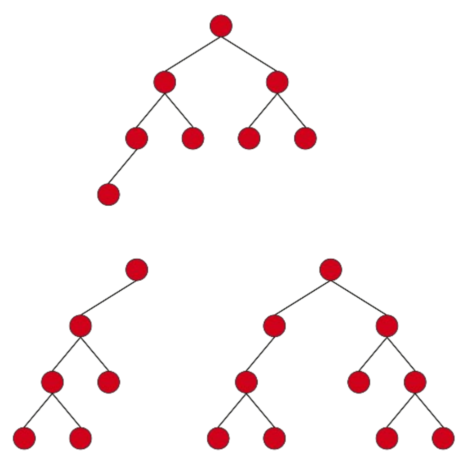


Full / Proper Binary Tree: Strictly binary. Every node has two children or none at all.

Full / Proper

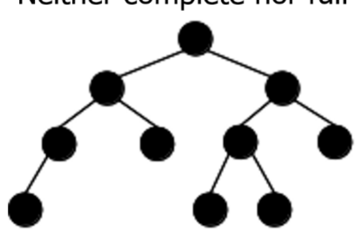


Not full / Improper

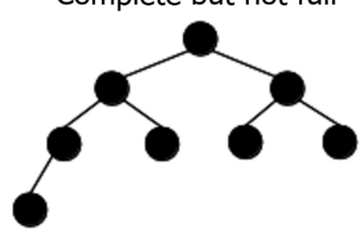


Complete vs. Full trees

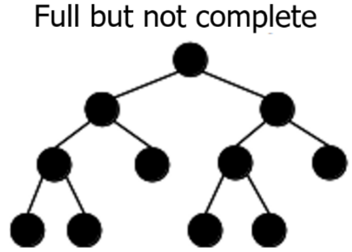
Neither complete nor full



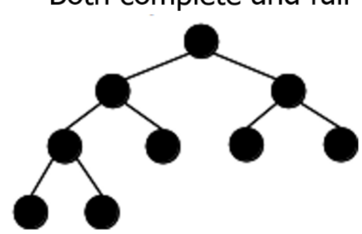
Complete but not full



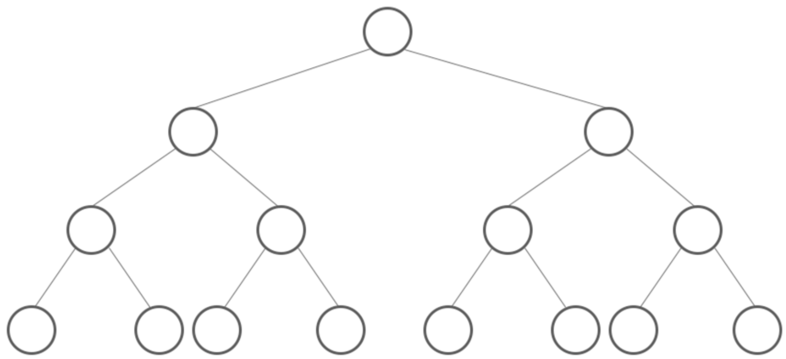
Full but not complete



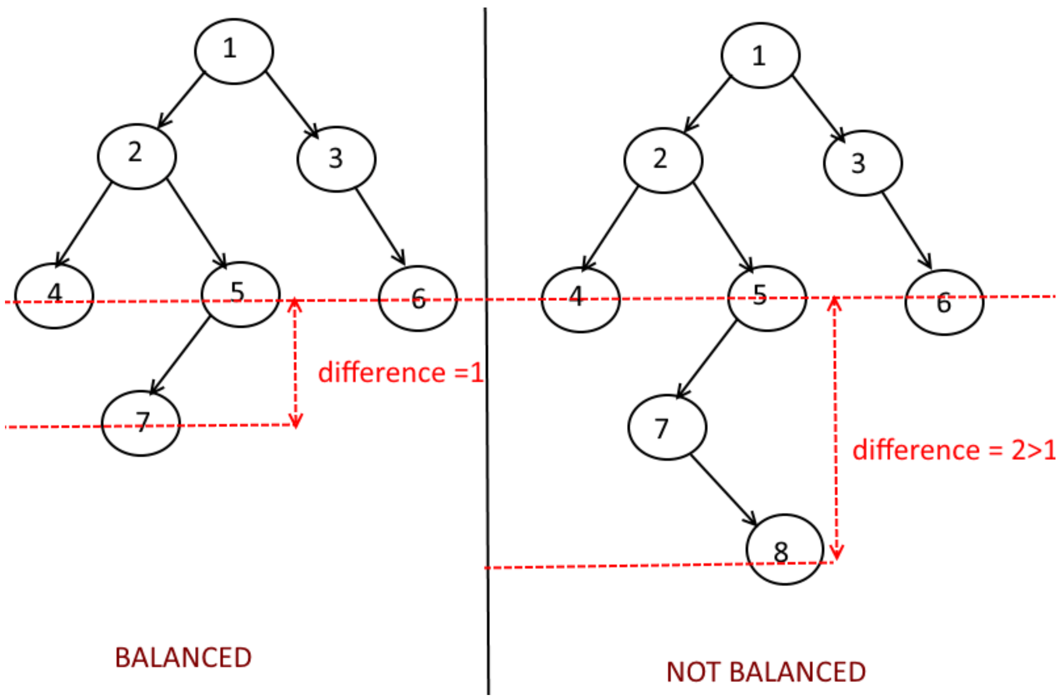
Both complete and full



Perfect: A binary tree is a complete binary tree with the last level also filled completely.



A **balanced binary tree** is a binary tree structure in which the left and right subtrees of every node differ in height by no more than 1.



Degenerate: A tree where each parent node has only one associated child node. This means that performance-wise, the tree will behave like a linked list.